

Non-Invasive Hemoglobin Scanner

A compact, battery-powered optical device that estimates blood hemoglobin from a fingertip — built for low-cost, point-of-care anemia screening.

MCU · Nordic nRF54L

Sensor · MAX30102

Firmware · Zephyr RTOS

Output · Hb in g/dL

01 Overview

The scanner measures blood hemoglobin without a needle. It shines red and infrared light into a fingertip, reads how much light is reflected back, and converts that signal into a hemoglobin estimate. The full system — sensor, processor, power, and display — is built as a small, battery-powered unit so it can be used for quick anemia screening outside a laboratory.

02 System architecture

Four building blocks make up the device. A Li-Po battery feeds the **nPM1300** power-management IC, which conditions and protects the supply for the **nRF54L** microcontroller. The microcontroller reads the **MAX30102** optical sensor over I²C and drives the on-device screen, all coordinated by firmware running on the **Zephyr** RTOS.

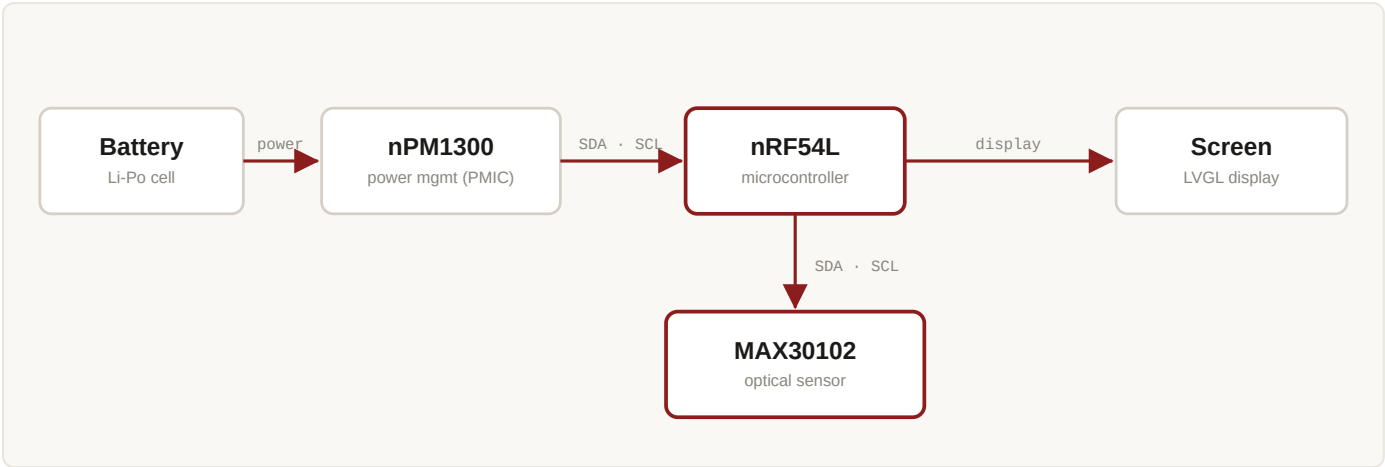


Figure 1 — Signal and power flow. The nPM1300 also supplies the regulated power rails to the microcontroller; SDA · SCL denote the shared I²C bus.

03 Components

PART	WHAT IT DOES	KEY DETAIL
nRF54L Nordic · SoC MCU	The brain. Runs the firmware, reads the sensor, and drives the display.	Arm Cortex-M33 @ 128 MHz · I ² C / SPI / UART · ~0.7 µA in sleep
nPM1300 Nordic · PMIC	Charges and protects the battery, regulates the supply rails, and reports battery state.	32-800 mA charging · ~370 nA quiescent · I ² C control
MAX30102 Analog Devices · sensor	The sensing front-end. Emits red and IR light into the fingertip and measures the reflected light.	Red 660 nm + IR 880 nm · photodetector · I ² C
Zephyr real-time OS	Schedules tasks and manages the drivers so the application code stays simple.	Industry-standard embedded RTOS
LVGL display library	Renders the reading and charts on the device screen.	Open-source embedded GUI library
Li-Po battery power source	Rechargeable portable power, managed by the nPM1300.	Single-cell lithium-polymer
Display readout	Shows the hemoglobin estimate to the user on-device.	Driven via LVGL

04 Working principle

The hemoglobin estimate is produced in four stages, from raw light to a calibrated value.

- 1

Capture. The MAX30102 shines red 660 nm and infrared 880 nm light into the fingertip and records the reflected intensity (reflectance) at each wavelength.
- 2

Convert. The raw reflectance readings are converted into absorbance values.
- 3

Compute. The absorbance values are combined through matrix operations (matrix multiplication) to produce the derived measurement values.
- 4

Calibrate. Those values are mapped to a hemoglobin concentration using a linear calibration equation drawn from published research. The result is shown on the screen as an estimate in g/dL.

05 Clinical validation

The prototype was tested on subjects during hospital visits. Each device estimate is compared against a reference hemoglobin value below (g/dL).

SAMPLE	DEVICE EST.	REFERENCE	Δ
1	15.50	15.50	0.00
2	13.20	12.62	0.58
3	14.10	14.60	0.50
4	13.50	13.51	0.01

Across these four samples, the device estimate differed from the reference by at most 0.58 g/dL, with a mean absolute difference of ≈ 0.27 g/dL — tracking closely with the reference values.

06 Proof of clinical testing

Measurements were taken at the hospital using the fingertip as the optical site for the reflectance sensor. The photographs below document two of the testing sessions.



Subject – Rama Darshni
Fingertip measurement site · Hospital Day 1



Subject – Amisha
Fingertip measurement site · Hospital Day 1